

Relativity

Building Models to Describe our World

History of Relativity, Einstein and Spacetime Phenomena



Outline

History of Relativity

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Outline

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- Causality, space-like, time-like

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History of Relativity

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History of Relativity, Einstein and Spacetime Phenomena

Maxwell's Equations

- So far we have seen several apparently disconnected topics in e/m:
 - Coulomb's law or Gauss' law (we showed that these are equivalent):

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q^{enc}}{\epsilon_0}$$

- Ampere's law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I^{enc}$$

- Faraday's law:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A}$$

- No magnetic monopoles (like Gauss' law, but no such thing as magnetic charge):

$$\oint \vec{B} \cdot d\vec{A} = 0$$

Maxwell's Equations

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 - Coulomb's law or Gauss' law (we showed that these are equivalent):

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q^{enc}}{\epsilon_0}$$

- Ampere's law (a little more generally):

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I^{enc} = \mu_0 \int \vec{j} \cdot d\vec{A} + \mu_0 \epsilon_0 \frac{d}{dt} \int \vec{E} \cdot d\vec{A}$$

- Faraday's law:

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A}$$

- No magnetic monopoles (like Gauss' law, but no such thing as magnetic charge):

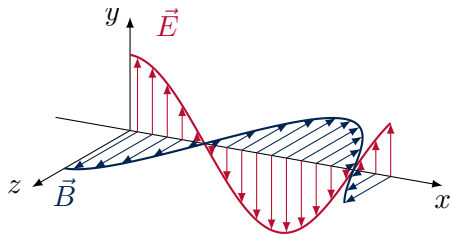
$$\oint \vec{B} \cdot d\vec{A} = 0$$

→ $\vec{B}$ is related to a time-varying $\vec{E}$, and vice versa.

Maxwell's equations and electromagnetic waves

- Maxwell realized that we can combine the 4 equations together into a unified model of electricity and magnetism, ~1860.
- An example of “unifying” forces in physics (electric and magnetic).
- As a consequence of Maxwell's equations, we have electromagnetic waves that can propagate. Maxwell realized that light was such a wave.
- The speed of those waves is:

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$



Electromagnetic wave propagating in the x direction.

From Maxwell's equations:

$$\begin{aligned}\frac{\partial^2 E_y}{\partial t^2} &= \mu_0 \epsilon_0 \frac{\partial^2 E_y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 E_y}{\partial x^2} \\ \frac{\partial^2 B_z}{\partial t^2} &= \mu_0 \epsilon_0 \frac{\partial^2 B_z}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 B_z}{\partial x^2}\end{aligned}$$

Electromagnetic waves

- Thanks to Maxwell, physicists in the late 1800s realized that light is an electromagnetic wave that propagates with speed:

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c = 3 \times 10^8 \text{ m/s}$$

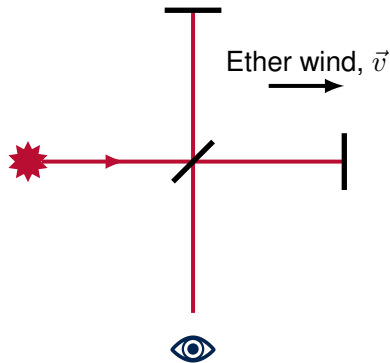
- But...
 - What is the medium in which the light waves propagate with that speed?
 - Presumably, that speed is then relative to that medium.

The ether

- It was postulated that there exist a medium, the “ether”, in which electromagnetic waves propagate.
- That ether must exist everywhere around us, including in vacuum.
- It must be some sort of “absolute reference frame” in the Universe through which we move.
- So physicists, Michelson and Morley, decided to see if we can measure the speed of Earth through the ether, as it moves around the Sun with a speed 30 km/s (3×10^4 m/s) or 1 part in 10 000 of the speed of light.

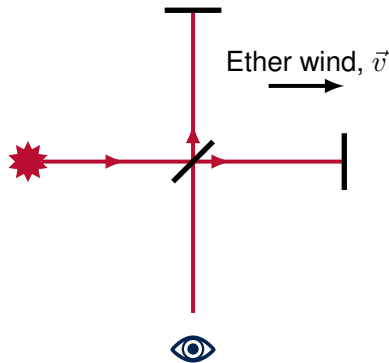
The Michelson-Morley experiment

- A source of light is split into 2 perpendicular paths.



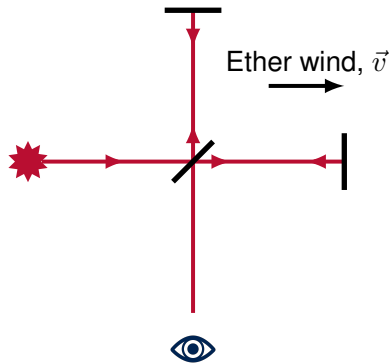
The Michelson-Morley experiment

- A source of light is split into 2 perpendicular paths.
- If relative speed through ether is 0, it will take the same amount of time for the light to go along each path.
- Look for any difference in time by inspecting the interference pattern of the light waves.
- Rotate the apparatus to eliminate most uncertainties.



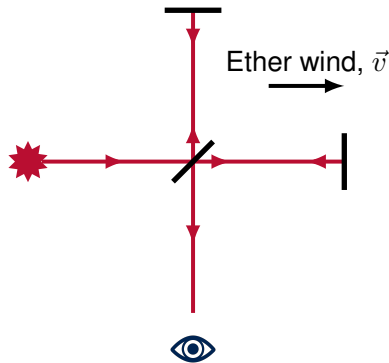
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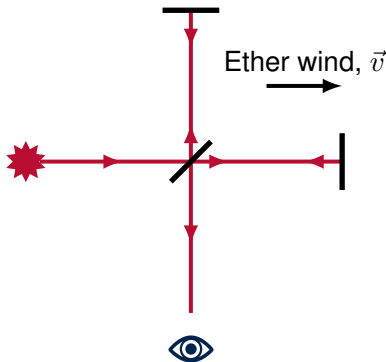
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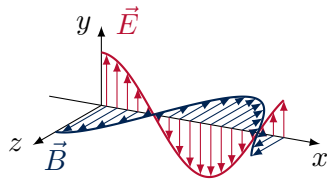


Null result from Michelson and Morley

- After several years of refinements, the Michelson and Morley experiment was not able to detect any effect from motion through ether.
- The light took the same amount of time to go in either leg of the interferometer.
- This was very perplexing.

Einstein's thought experiment

- Albert Einstein is now famous for his “thought experiments”
- **Unaware of Michelson and Morley's null result**, Einstein asked himself what the electric and magnetic fields would look like if he was moving along with an electromagnetic wave.
- He would then see E and B fields that oscillate in space, but not in time:
 - There is no way to put this into Maxwell's equations and have a consistent distribution of charges/currents to give such fields
- **In a frame of reference moving at the speed of light, Maxwell's equations do not work.**



Relativity

- Galileo and Newton both realized that mechanical laws of physics are independent of the reference frame (as long as all reference frames are inertial, non-accelerating).
 - In other words, there is no experiment that you can do to measure your “absolute” speed, only your relative speed.

Relativity

- Galileo and Newton both realized that mechanical laws of physics are independent of the reference frame (as long as all reference frames are inertial, non-accelerating).
 - In other words, there is no experiment that you can do to measure your “absolute” speed, only your relative speed.
 - If you observe someone doing an experiment in a train moving relative to you at constant velocity, you will both agree that $F = ma$, and you will agree on the values of F , m , and a .
 - If the person in the train drops a mass, you will agree on how long it takes for it to hit the ground, the forces on it, its acceleration, etc.

Relativity and Maxwell's equations

- Einstein had a choice:
 - Either Maxwell's equations are not the same in different reference frames .
 - There is something funny going on.
- He chose the latter, and postulated (the first postulate):

All the laws of physics ($F=ma$, Gravity, Maxwell's equations) are the same in all inertial reference frames. There is no special reference frame for Maxwell's equations to work. There is no way to measure your absolute speed, no absolute frame of reference, no ether.

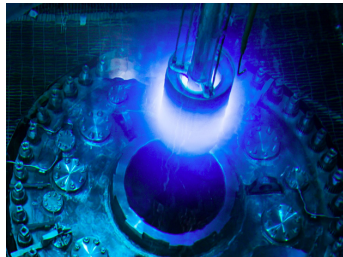
Einstein's second postulate

- In order to resolve his thought experiment, Einstein needed a second, dramatic postulate:

The speed of light in vacuum is c . No matter what reference frame you measure it in. It doesn't depend on the speed of the light source or of the observer. You cannot be in a frame of reference where light appears immobile or to have a speed that is not c .

The speed of light in vacuum

- Einstein's postulate concerns the speed of light in vacuum, c , when it travels without interacting with matter.
- In a medium, light travels slower than c (recall, refraction, $v = cn$).
- It is possible for particles in a medium to travel faster than the speed of light in that medium (Cherenkov radiation), but not faster than c (kind of like a sonic boom, but with light!).



Electrons emitted in the core of a nuclear reactor travel faster than the speed of light in water, causing blue Cherenkov radiation



Should be blue/purple, not green. Also, fission not enough to produce significant Cherenkov radiation in air...

Consequences of Einstein's postulates

- They explain the Michelson and Morley experiment, since you cannot expect that the light would take different amounts of time to go through the two legs of the interferometer.
- But they have a lot of much weirder consequences... that have been experimentally verified to a high degree of accuracy!

Results and Examples of Einstein's Relativity

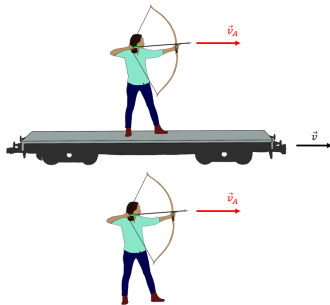
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Review: Einstein's Postulates

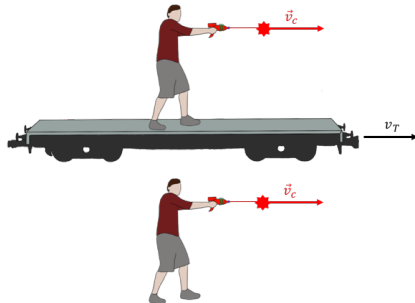
1. All the laws of physics ($F = ma$, Gravity, Maxwell's equations) are the same in all inertial reference frames.
 - *There is no special reference frame for Maxwell's equations to work. There is no way to measure your absolute speed, no absolute frame of reference, no ether.*
2. The speed of light is always measured to be c , in all inertial reference frames.
 - *It doesn't depend on the speed of the light source or of the observer, everyone measures c .*

The speed of light is *not* relative



In **Classical Physics**, velocities add up:

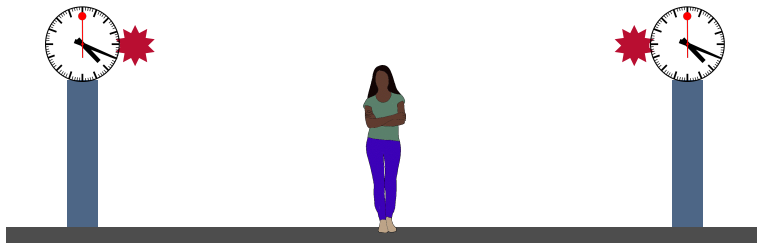
- The arrow from the train travels with speed $v = v + v_A$ relative to the ground.



In **Special Relativity**, everyone measures c for the speed of light

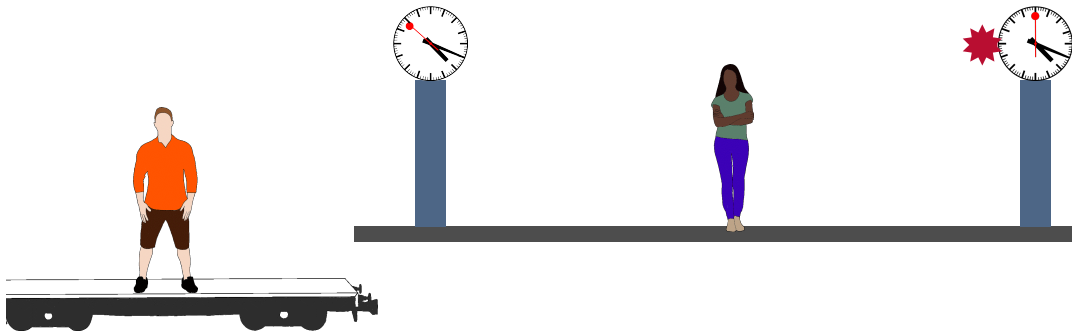
- The light from the train travels with speed $v = c$ relative to the ground.

Simultaneity



- An observer on a platform is equidistant from 2 clocks that output pulses of light. When the clocks read 16:20, they output a light pulse, the observer sees the two light pulses at the same time and she concludes that the clock are synchronized.

Simultaneity



- An observer on a train sees the platform moving in his rest frame. He observes two pulses of light arrive at the same time at the person on the platform.
- He concludes that the pulse from the right happened first, since the pulses travel at speed c and the one from the right travelled further. He disagrees that the clocks are synchronized! **But, causality...**

Defining “events” and frames of reference

- We considered the “event”: *“The person on the platform received two pulses of light at the same time.”*
- We did not consider the “event”: *“The two clocks emitted a pulse of light at the same time.”*
- This led to the conclusion that simultaneity is relative:
 - The second statement is not an “event” because the two clocks are at different positions in space.
 - We can only compare things that happen at the same location in space and time. An “event” is something that occurs at a single location in space and time.
 - We introduce “space-time”, and we say that an event is something that happens in four-dimensional space-time and has coordinates (x,y,z,t)
 - We can then compare if events in space-time are simultaneous or not:
 - “Person receiving two pulses of light” is always after “clock A” and “clock B” emit a pulse of light. You can check this out; the only way for this not to be true is if one of the observers goes faster than c .
 - “clock A emits a pulse of light” is not always at the same time as “clock B emits a pulse of light”. We just showed that it can be before or after as well!

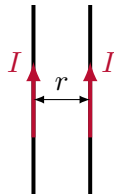
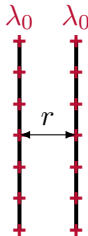
Example question

- Repulsive force per unit length between two charged wires:

$$\frac{F_0^E}{l} = \frac{\lambda_0}{2\pi\epsilon_0 r}$$

- Attractive force per unit length between two current carrying wires:

$$\frac{F_0^B}{l} = \frac{\mu_0 I^2}{2\pi r}$$



Example question

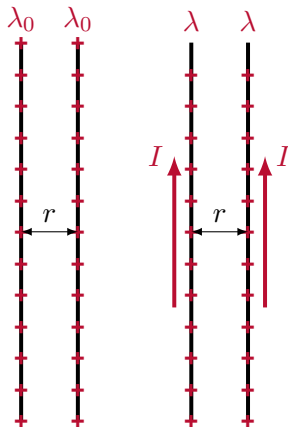
- In a frame moving downwards with speed v , it looks like there is an upwards current given by:

$$I = \frac{dQ}{dt} = \frac{d(\lambda l)}{dt} = \lambda v$$

- Let us assume that the laws of physics and that the constants of nature remain the same in this frame of reference. The force on one of the wires must remain the same:

$$\frac{F_0^E}{l} = \frac{\lambda_0}{2\pi\epsilon_0 r} = \frac{\lambda}{2\pi\epsilon_0 r} - \frac{\mu_0 I^2}{2\pi r}$$

Charge density is the only thing that can be different in the moving frame, so we call it λ .



Two charged wires viewed at rest (left) and from a frame of reference that is moving downwards (right).

Example question

Working through the math:

$$\frac{\lambda_0}{2\pi\epsilon_0 r} = \frac{\lambda}{2\pi\epsilon_0 r} - \frac{\mu_0(\lambda v)^2}{2\pi r}$$

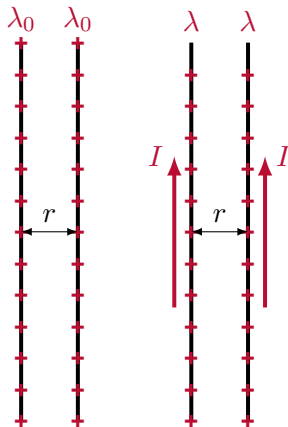
$$\frac{\lambda_0}{\epsilon_0 r} = \frac{\lambda}{\epsilon_0} - \mu_0(\lambda v)^2$$

$$\lambda_0^2 = \lambda^2(1 - \epsilon_0\mu_0 v^2)$$

$$\therefore \lambda = \lambda_0 \frac{1}{\sqrt{1 - \epsilon_0\mu_0 v^2}}$$

$$\boxed{\therefore \lambda = \lambda_0 \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}}$$

From Maxwell's equations: $c = \frac{1}{\sqrt{\epsilon_0\mu_0}}$



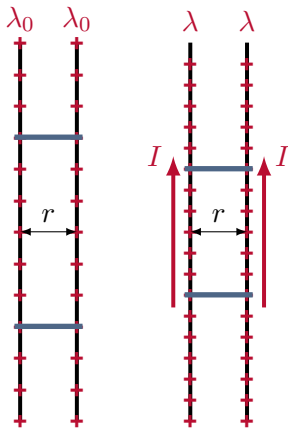
Two charged wires viewed at rest (left) and from a frame of reference that is moving downwards (right).

Example question

The charge density *is* different in the moving frame of reference (and larger since $v < c$):

$$\lambda = \lambda_0 \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

- Observers in the two reference frames can agree on the number of charges in a section of wire (they could literally count the charges between 2 lines drawn on the wire).
- The only way for the charge density to be higher in the moving reference frame is if **the wire is actually shorter (the lines are closer)!**

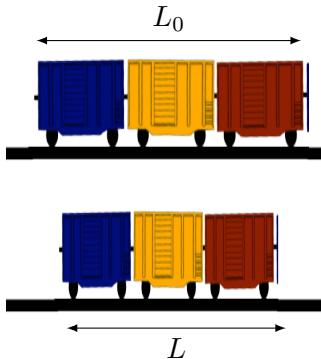


Two charged wires viewed at rest (left) and from a frame of reference that is moving downwards (right).

Length contraction

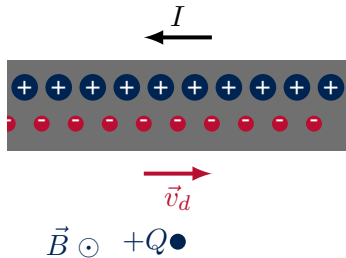
- An object that is moving is shorter in the direction of motion than you would measure if the object were at rest.
- This is real, stuff shrinks lengthwise as it moves fast.

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}} < L_0$$



Special Relativity and E/M

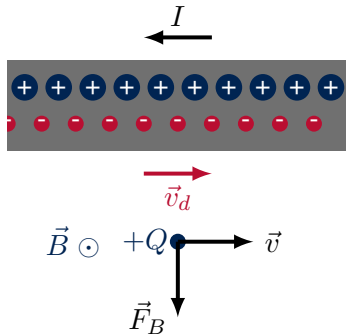
- A (neutral) wire carries current (electrons moving to the right relative to stationary ions).
- We place a charge $+Q$ near the wire, it feels no net force.
- We would say the current creates a magnetic field out of the page, but since Q is not moving, it feels no force ($\vec{F} = Q\vec{v} \times \vec{B}$).



In the reference frame of the wire: A charge is near a wire with current. The charge feels a magnetic force if it moves relative to the wire.

Special Relativity and E/M

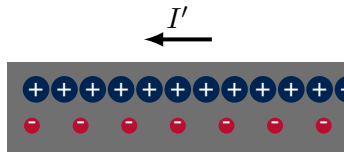
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- We would say the current creates a magnetic field out of the page, but since Q is not moving, it feels no force ($\vec{F} = Q\vec{v} \times \vec{B}$).
- If Q moved towards the right, it would feel a downward magnetic force.



In the reference frame of the wire: A charge is near a wire with current. The charge feels a magnetic force if it moves relative to the wire.

Special Relativity and E/M

Suppose that Q moves to the right, *with the same speed as the electrons*. **In the frame of Q :**



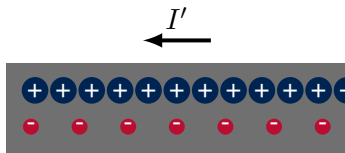
$$\vec{B}' \odot +Q \bullet$$

In the reference frame of the charge moving with the electrons.

Special Relativity and E/M

Suppose that Q moves to the right, *with the same speed as the electrons*. **In the frame of Q :**

- The electrons are immobile and *are sparse* (they were length contracted).
- The ions move to the left and *are denser* (they are length contracted).



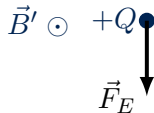
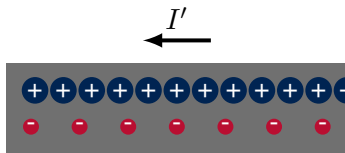
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In the reference frame of the charge moving with the electrons.

Special Relativity and E/M

Suppose that Q moves to the right, *with the same speed as the electrons*. **In the frame of Q :**

- The electrons are immobile and *are sparse* (they were length contracted).
 - The ions move to the left and *are denser* (they are length contracted).
- The wire no longer appears neutral and repels Q by the **electric force**, which must have the same magnitude.
- Since Q is stationary, there is no force from the magnetic field due to the moving ions.
- The magnetic and electric force are really different aspects of the same effect, Special Relativity is the connection!



In the reference frame of the charge moving with the electrons. It feels a purely electric force due to the apparent net charge on the wire.

Einstein's Clock

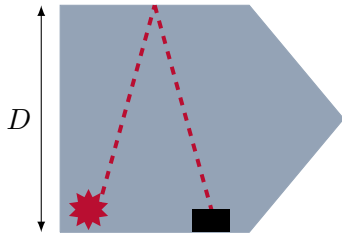
- A beam of light bounces back and forth between 2 mirrors.
- You watch the light beam, and determine that the time for a return trip of the light is:

$$\Delta\tau = \frac{2D}{c}$$

The speed of light is:

$$c = \frac{2D}{\tau}$$

Note: You measure the time difference between two events at the same position, as far as you are concerned.

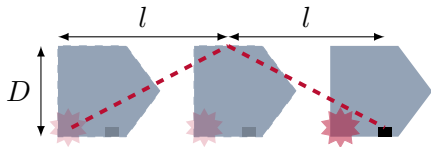


A beam of light can be used as a clock in this spaceship. Here the path of the beam is viewed in the reference frame of the spaceship.

Einstein's Clock

- You take your clock on a spaceship and someone on Earth observes the light beam.
- The light travels a longer distance as far as they are concerned, but the speed of the light beam is still c (in the diagonal direction in the above diagram).
- They measure the time difference between events that happen at different locations.
- They would write the speed of light as:

$$c = \frac{2\sqrt{D^2 + l^2}}{\Delta t} = \frac{2\sqrt{D^2 + \left(\frac{v\Delta t}{2}\right)^2}}{\Delta t}$$

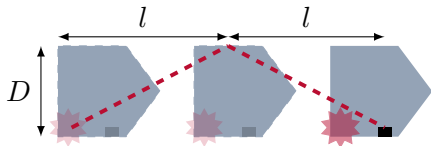


A beam of light can be used as a clock in this spaceship. Here, the path of the beam is viewed by someone on Earth, as the spaceship moves to the right with speed, v .

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A beam of light can be used as a clock in this spaceship. Here, the path of the beam is viewed by someone on Earth, as the spaceship moves to the right with speed, v .

Re-arranging to isolate time, Δt :

$$\Delta t = \frac{2D}{c\sqrt{1 - \frac{v^2}{c^2}}} = \frac{\Delta\tau}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Gamma

- The factor:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

comes up a lot, so we call it γ . v is the relative speed between two observers in inertial frames of reference.

Time dilation

$$\Delta t = \gamma \Delta \tau = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \Delta \tau$$

- The “**proper time**”, $\Delta \tau$, is the time that someone measures in their own reference frame (e.g. moving with the clock).
 - Δt is the time that would be measured in a frame that is moving with speed v relative to the frame where $\Delta \tau$ is measured.
 - Because $\gamma > 1$, $\Delta t > \Delta \tau \rightarrow$ “**Time dilation**”.
- The time measured in the moving reference frame is always longer than the proper time that is measured in a stationary frame.
- This is not a thought experiment, time actually goes by slower, it's experimentally verified.

Muons

- Muons are particles that radioactively decay in $2.2 \mu\text{s}$ ($2.2 \times 10^{-6} \text{ s}$).
- They are produced when cosmic rays hit our atmosphere, at about 10km of altitude.
- If they travel at essentially the speed of light, then to reach the Earth's surface, it takes them:

$$t = \frac{d}{v} \sim \frac{10 \text{ km}}{c} = 3 \times 10^{-5} \text{ s}$$

→ 10x times their lifetime, so we should see very few muons at the Earth's surface... but we see plenty (about 100,000 more than we'd expect!).



Illustration of cosmic rays generating muons in the atmosphere. <http://twinparadox.net/>

Time dilation *implies* length contraction

- Since you saw the Earth move away at speed $\sim c$ for about 4.5 years before you got to the star, you conclude that the star is about 4.5 light years from Earth.
- You experience length contraction:

$$L = \frac{L_0}{\gamma}$$

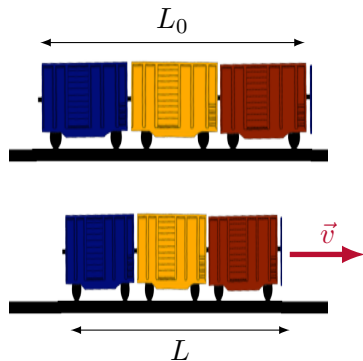
→ The proper length, L_0 , is the length as measured in a frame at rest with respect to the object.

If you are at rest with the Earth and star, you measure that they are $L_0 = 100$ light years apart. If you move with respect to them, you measure a shorter distance, L .

Relativistic train

- You are moving with the train, so you measure the proper length of the train at rest.
- People on Earth see the train moving, so they see it being contracted.

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}} < L_0$$



Train and tunnel

- 200 m is the proper length of the tunnel as measured at rest on Earth. From the train, you see it being shorter:
 - Your train has a proper length of 500 m, and is measured to be 200 m on Earth, so:

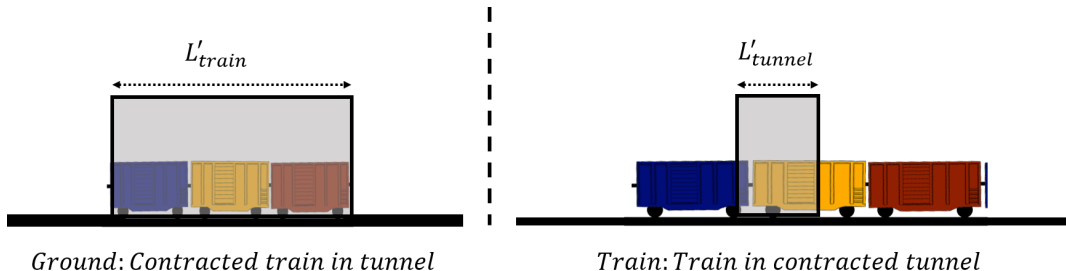
$$L = \frac{L_0}{\gamma} \rightarrow 200 \text{ m} = \frac{500 \text{ m}}{\gamma} \rightarrow \gamma = 2.5$$

- In your reference frame, you measure the tunnel to be:

$$L = \frac{L_0}{\gamma} = \frac{200 \text{ m}}{2.5} = 80 \text{ m}$$

→ So you really don't think your 500 m train fits in the 80 m tunnel!

The Train Paradox



- As seen from the ground, the train that is contracted to a length of 200 m fits in the tunnel of the same length. As seen in the train, the 500 m train does not fit in the 80 m tunnel.
- A person on Earth briefly closes the doors of the tunnel while the train is inside, nothing happens to the train, since it fits. To the observer on the train, was the train just destroyed?

Simultaneity

- Remember, the people on Earth and on the train do not agree on the timing of events!
- If the person on Earth closes the tunnel when the locomotive reaches the exit, the person on the train can agree that the exit door closed just before the locomotive reached the exit → Event A!
- If the person on Earth closes the tunnel just after the last car entered the tunnel, the person on the train can agree that the entrance door closed just after the last car entered the tunnel → Event B!
- To the person on Earth, A and B are simultaneous.
- To the person the train, A and B are not simultaneous.
- Question, to the person on the train, which happened first, A or B?

The Math and Geometry of Spacetime

Building Models to Describe our World

History of Relativity, Einstein and Spacetime Phenomena

Converting time and length

- So far, have seen how to convert intervals of length and time from one frame of reference to another.
- Time dilation:

$$\Delta t = \gamma \Delta \tau$$

- Length contraction:

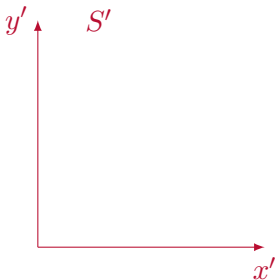
$$L = \frac{L_0}{\gamma}$$

- Lorentz factor:

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

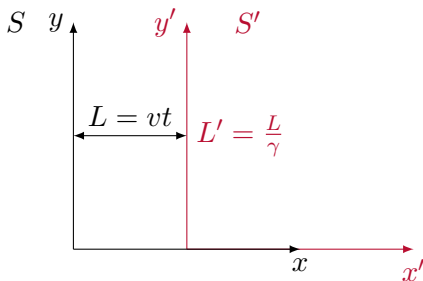
Lorentz Transformations

- More generally, want to be able to convert any coordinates of an “event” between frames of reference, S and S' moving with velocity $\vec{v}$ relative to each other.
- By convention, we choose the x -axis in both frames to be parallel to $\vec{v}$, and the origins are coincident at $t = 0$:
 - Only get length contraction along the x direction.



Lorentz Transformations

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In Gallilean relativity

One would write:

$$L = L'$$

$$vt = vt'$$

which cannot be consistent with SR...

Lorentz Transformations

- In order to get results that are consistent with SR, we cannot just transform the spatial coordinates (x, y, z) between frames of reference; we also need to change the time between coordinate systems!
- Remember, time dilation *implies* length contraction and vice versa. If we change spatial coordinates between frames, we have to change time as well, or observers would disagree on the speed of light.
- We must thus use 4 coordinates to describe an event in a frame of reference $\rightarrow$ we have to treat time as another coordinate. (x, y, z, t) and (x', y', z', t') .
- We thus introduce the notion of “Space-time” as a 4D coordinate system where time is on “equal footing” with the spatial coordinates.

The Lorentz transformations

$$x' = \gamma(x - vt)$$

$$y' = y$$

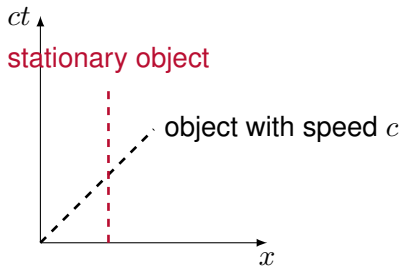
$$z' = z$$

$$t' = \gamma \left(t - \frac{vx}{c^2} \right)$$

When v is small ($\gamma \sim 1$), one recovers Galilean relativity ($x' = x - vt$, $t' = t$).

Space-time diagrams

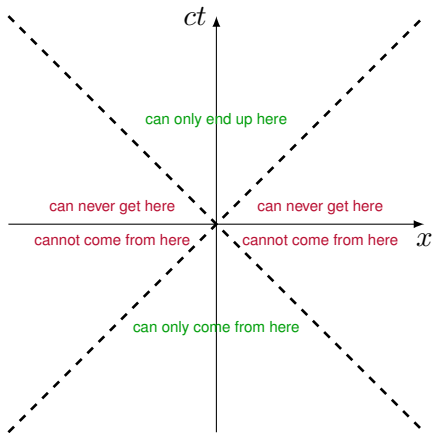
- It's hard enough to draw 3D diagrams, so 4D diagrams are even harder...
- To help us visualize Space Time, we draw space time diagrams showing only x and t coordinates (think of showing displacement vs time, except that this is *position and time*).
- Instead of t on the vertical axis, we multiply by c to get dimension of length.
- An object's trajectory in space time is called its "world line":
 - A vertical world line corresponds to a stationary object (t keeps increasing while x is constant).
 - A line at 45 degrees corresponds to moving at the speed of light ($ct = x$).



Space-time diagram.

Light cones

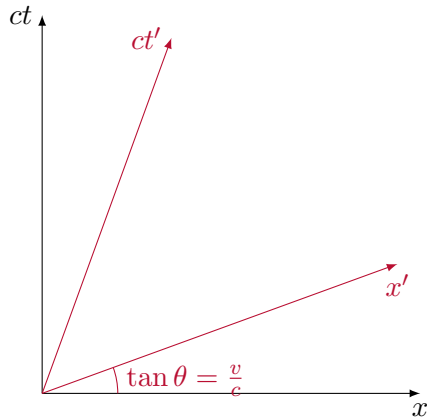
- Since no signal can travel faster than the speed of light, there is always a limit as to where an object could have been in the past and where it could be in the future.
- If an object is at the origin of a coordinate system at $t = 0$, we can draw 2 cones that determine where the object could have been in the past and where it could be in the future.



Space-time diagram with light cones.

Space-time diagrams in different frames

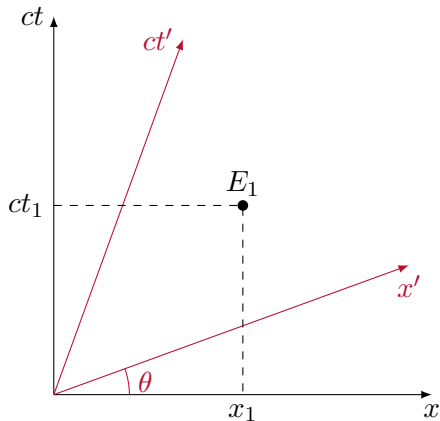
- In space time, we can draw S and S' with the same origin.
- However, because of the Lorentz transformation, the (x, ct) axes of S' are not perpendicular as viewed in S .
- Suppose an event, E_1 , occurs in space-time:



Space-time diagram in two different reference frames

Space-time diagrams in different frames

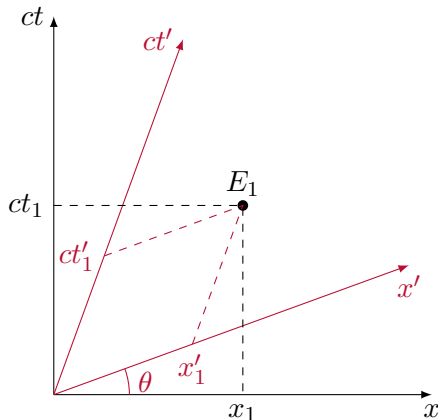
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 - In S , it has position (x_1, ct_1) .



Space-time diagram in two different reference frames

Space-time diagrams in different frames

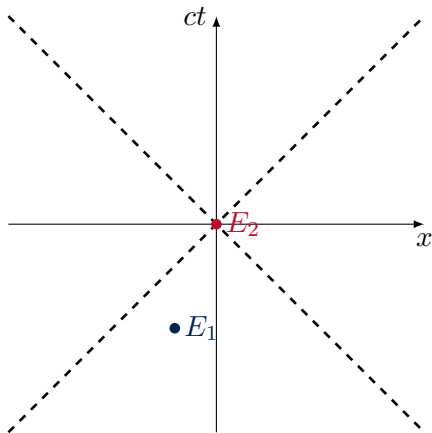
- In space time, we can draw S and S' with the same origin.
- However, because of the Lorentz transformation, the (x, ct) axes of S' are not perpendicular as viewed in S .
- Suppose an event, E_1 , occurs in space-time:
 - In S , it has position (x_1, ct_1) .
 - In S' , it has position (x'_1, ct'_1) given by Lorentz transformations.



Space-time diagram in two different reference frames

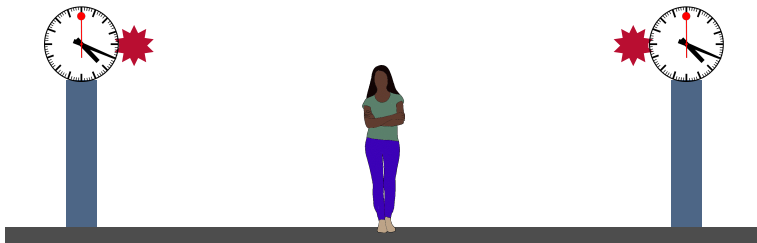
Causality, space like, time like

- One event can only cause the other if it is in the past light cone: there is no reference frame where it could look like E_2 happened before E_1 . These are called “**time like**”.
- Two events that are outside each other's light cones are called “**space like**” – they can never be causally connected. *These are the only types of events where observers can disagree on the order!*



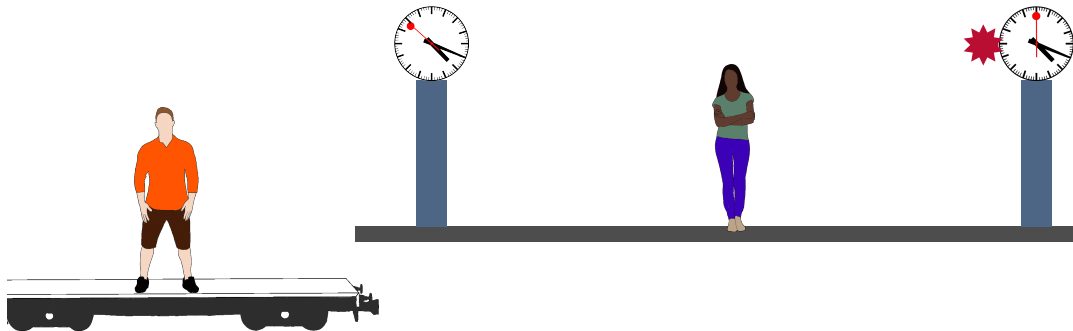
Space-time diagram showing time like events.

Recall: Simultaneity



- An observer on a platform is equidistant from 2 clocks that output pulses of light. When the clocks read 16:20, they output a light pulse, the observer sees the two light pulses at the same time and she concludes that the clock are synchronized.

Recall: Simultaneity



- An observer on a train sees the platform moving in his rest frame. He observes two pulses of light arrive at the same time at the person on the platform.
- He concludes that the pulse from the right happened first, since the pulses travel at speed c and the one from the right traveled further. He disagrees that the clocks are synchronized!

Space-time diagram for platform clocks

Draw the space time diagram for the case of the person on the platform in the middle of the two clocks:

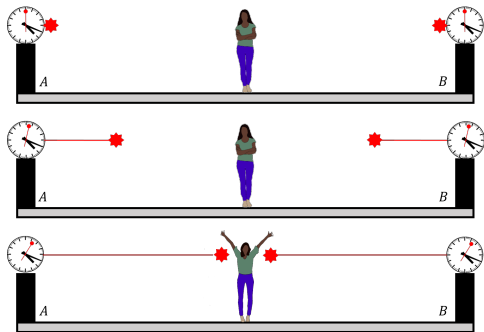
- Event A: Clock A emits a pulse of light
- Event B: Clock B emits a pulse of light
- Event C: The pulses of light meet at the position of the person

After drawing this in frame S (the platform), draw S' on top of that (the frame for the moving train), and compare the locations of A and B ...

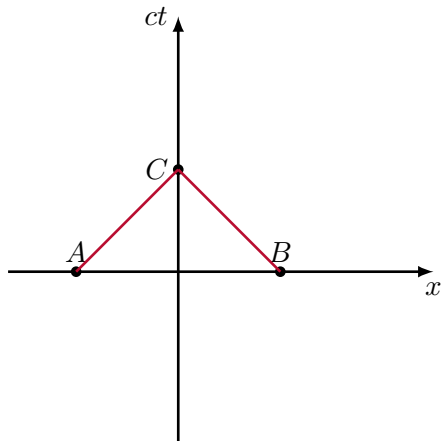


In the reference frame at rest relative to the clocks, they appear synchronized, emitting pulses of light at the same time.

Space-time diagram for platform clocks

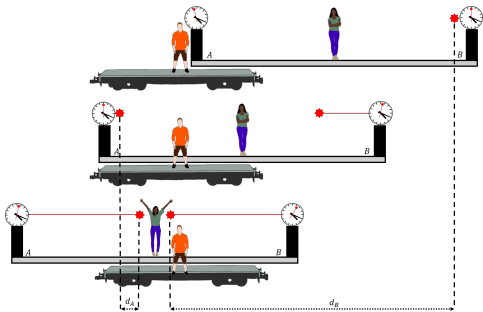


In the reference frame at rest relative to the clocks.

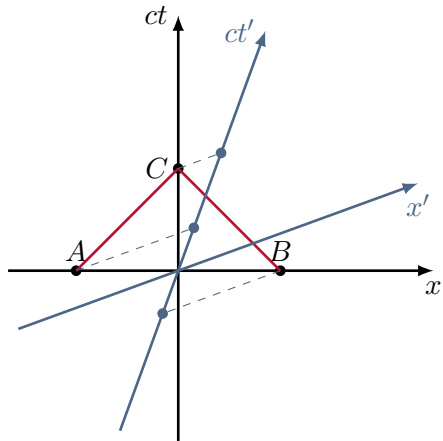


Space-time diagram for platform clocks

Space-time diagram for platform clocks

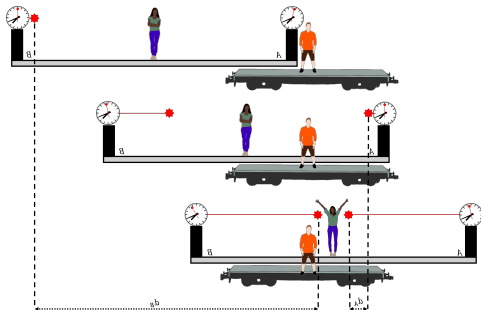


In the reference frame moving relative to the clocks.

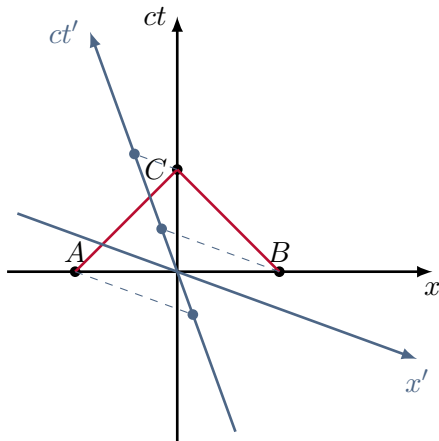


Space-time diagram for platform clocks

Space-time diagram for platform clocks

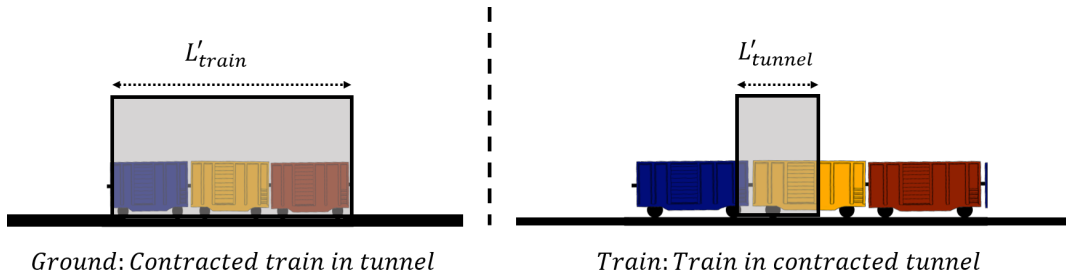


In the reference frame moving relative to the clocks, going the other way.



Space-time diagram for platform clocks

The Train Paradox



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